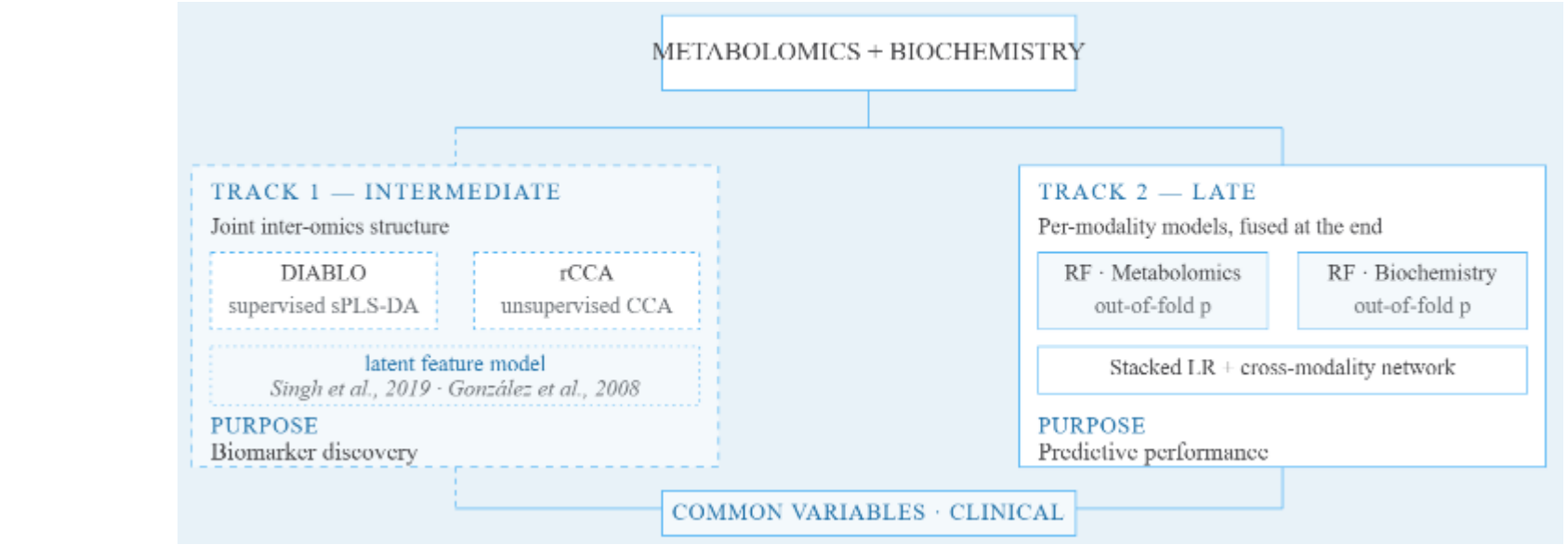
CRC MULTIMODAL INTEGRATION

Colorectal cancer (CRC) control analysis where biochemistry alone led prediction, but integration explained the biology.

RIYA DEEPAK SHET

TWO STRATEGIES

Same modality pair (metabolomics + biochemistry) compared on **intermediate discovery** and **late prediction**.

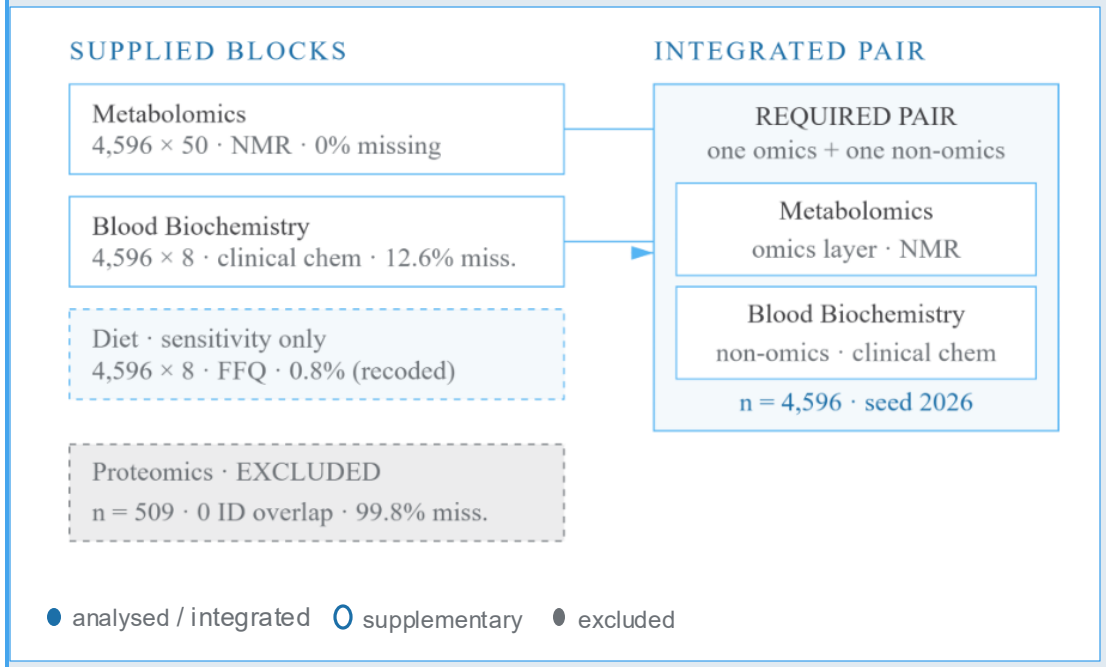


Why no early integration?
Concatenating 8-D biochemistry with 50-D NMR swamps biochemistry on dimensionality and scale (Picard et al., 2021). Intermediate preserves block structure; late preserves block learners.

DATA OVERVIEW

Matched samples determined what could be integrated.

- **n = 4,596,** balanced (Control 2,327 / CRC 2,269; 49.4% / 50.6%) → stratified CV, no resampling needed.
- Three supplied blocks share identical sample IDs; missingness is otherwise low and concentrated in biochemistry.
- Highest-missing biochemistry features:
  - Lipoprotein-A (24.0%)
  - Direct bilirubin (17.3%)
  - Apolipoprotein-A (12.9%)
  - HDL-cholesterol (12.4%)
- Diet retained as late-fusion sensitivity input. Proteomics excluded because patient-level integration requires matched IDs.

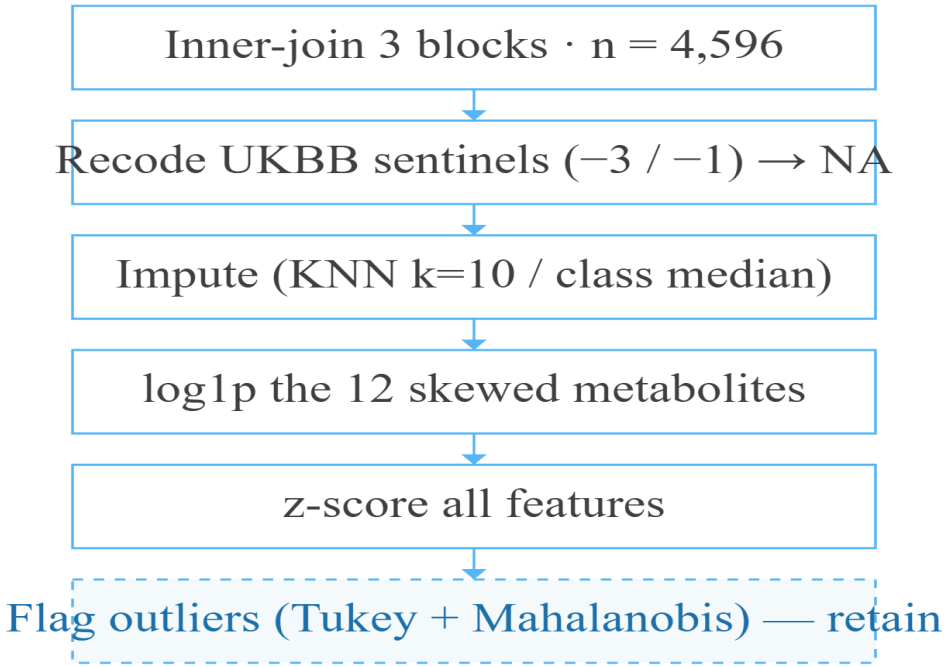


PREPROCESSING

Modality-matched cleaning; the strategy follows each block's structure.

- **Imputation, modality-matched:**
  - Metabolomics → KNN (k = 10) preserves correlation across collinear NMR subfractions (Troyanskaya et al., 2001).
  - Biochemistry → per-class median since low cardinality, clinically anchored ranges.
  - Diet → median after sentinel recoding.
- **UK Biobank sentinels recoded** (codes −3 / −1 → missing): 143 cells. Otherwise, they act as fake "negative intake".
- **log1p** on 12 / 50 right-skewed metabolites, then z-score every numeric feature.
- **Outliers flagged, not dropped** (Tukey 1.5 × IQR + Mahalanobis χ^2, $p < 0.001$); clinical features have legitimately long tails (Filzmoser et al., 2008).

PIPELINE

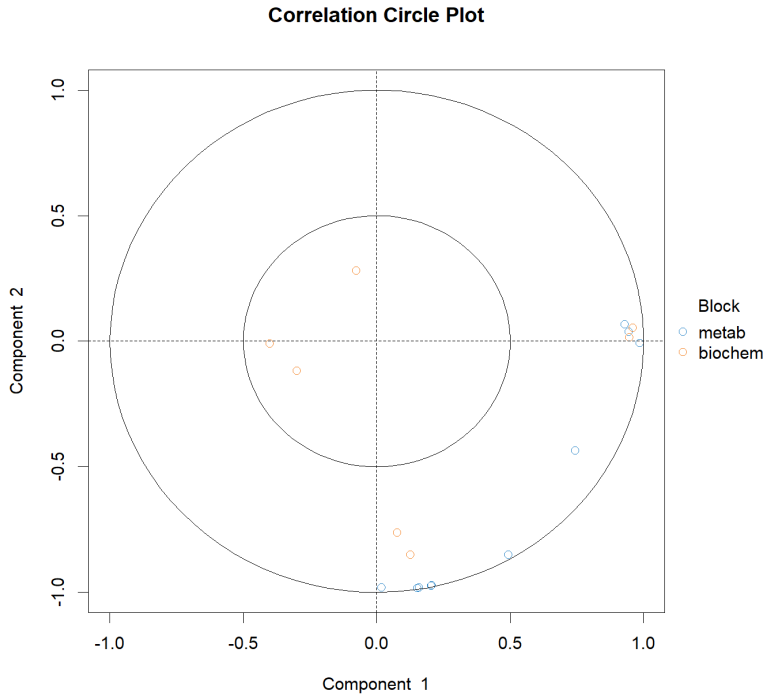


Modality-matched. Seed 2026 fixes every random draw.

INTERMEDIATE INTEGRATION

A strong joint inter-omics structure: two paradigms, one shared HDL / ApoA-I axis.

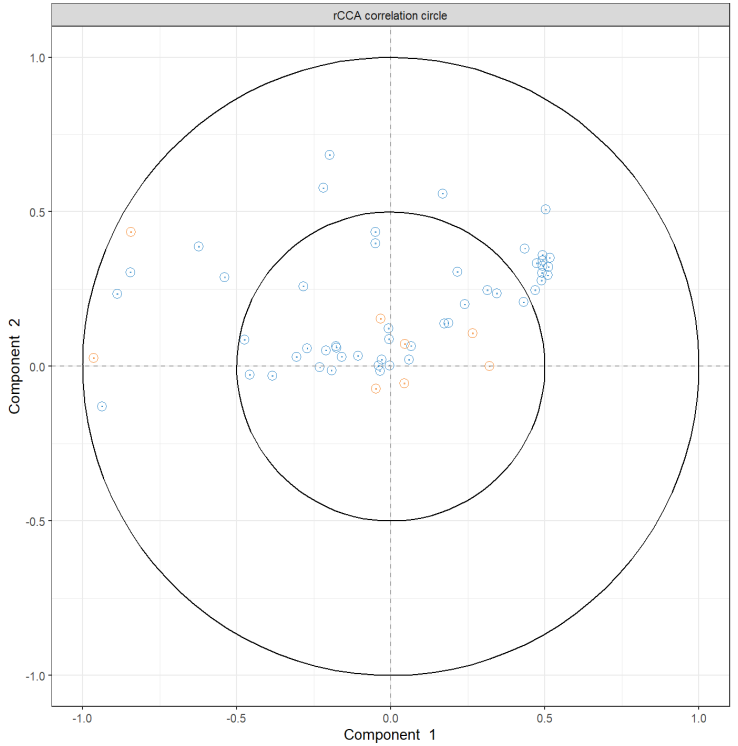
DIABLO · CORRELATION CIRCLE



outer ring = ±1 loading; inner = 0.5

Two symbol sets = the two blocks; variables far from the origin load most on the shared component.

RCCA · VARIABLE PLOT



Variables at the rim load most strongly on the canonical axis.

- **DIABLO** *(Singh et al., 2019; mixOmics)* recovers a sparse joint signature; component 1 loads on the **HDL / Apolipoprotein-A axis** in both blocks at once.
- **rCCA** (regularised CCA; González et al., 2008) component 1 canonical correlation **ρ = 0.884** between the blocks.
- Shared latent factor, the variables driving comp 1:
  - Metabolomics: Cholesterol in Large HDL, Total Lipids in HDL
  - Biochemistry: HDL-cholesterol, Apolipoprotein-A

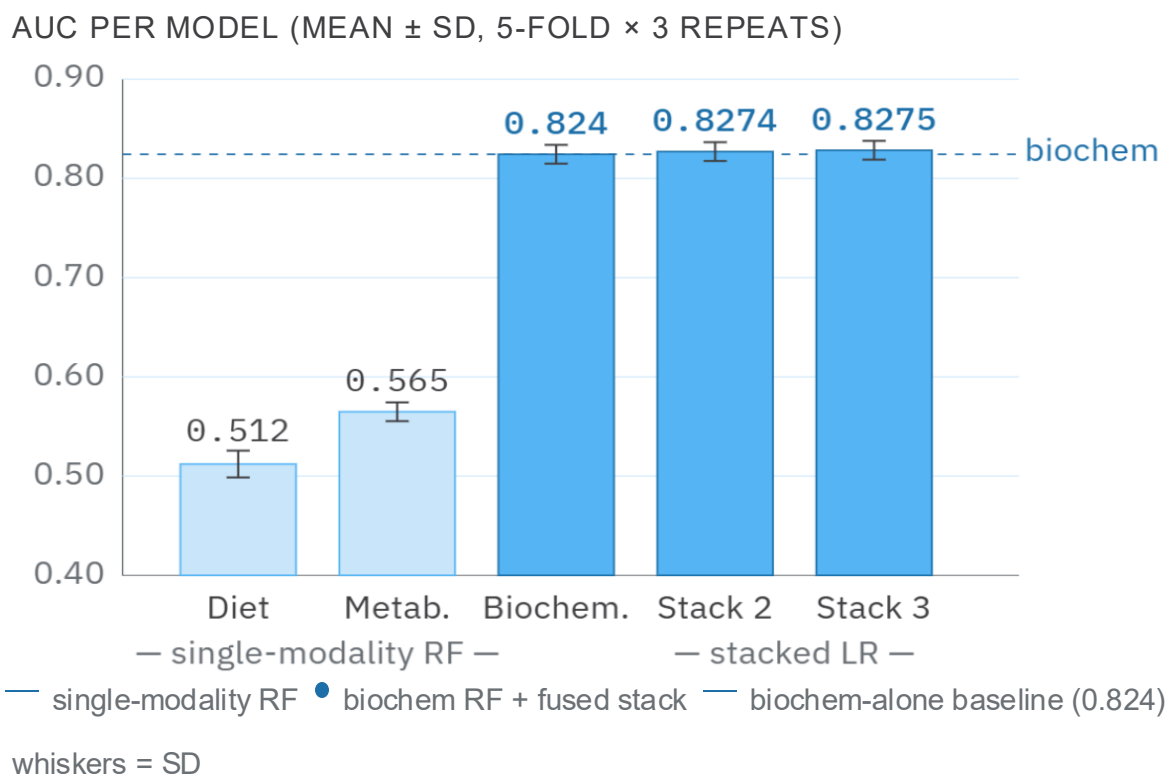
DIABLO with modest AUC 0.59.

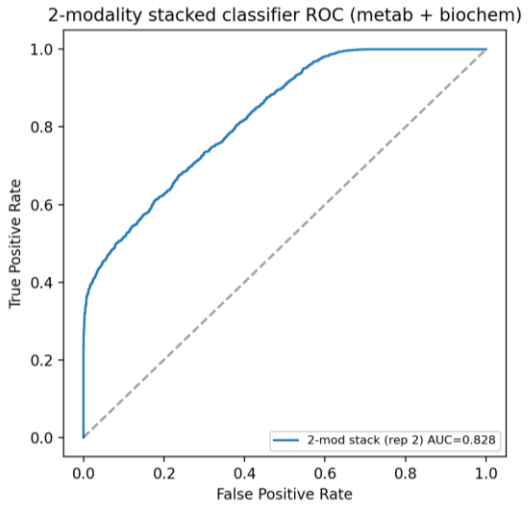
DIABLO is a latent feature model, not a classifier. Its sparse linear projection cannot match RF on prediction by construction. Its value is the interpretable joint loadings.

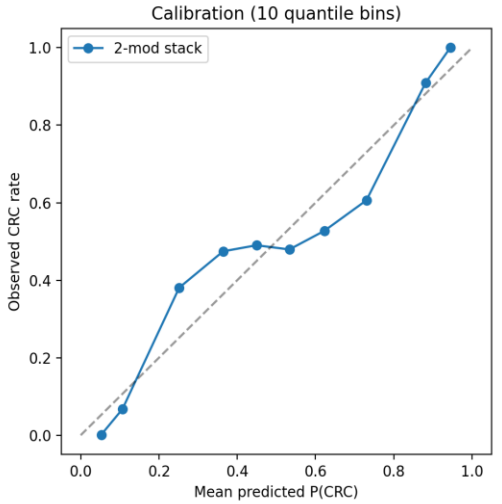
LATE INTEGRATION

Per-modality RFs → stacked LR. Fusion adds +0.0034 AUC over biochemistry alone.

- **Per-modality Random Forests:** biochemistry **AUC 0.824** +/- 0.010. Metabolomics and diet are weak standalone predictors.
- **Stacked LR fusion:** logistic regression on out-of-fold RF probabilities (leave-one-fold-out within each repeat).
  - metab + biochem = **AUC 0.8274** +/- 0.009.
  - Adding diet gives 0.8275; a **+0.0001** change.
- The **+0.0034 gain** is metabolomics *refining* the biochemistry decision. Calibration is acceptable (Brier 0.167).
- 5-fold × 3 repeats · stratified · seed 2026 · RF mtry tuned by inner-CV AUC



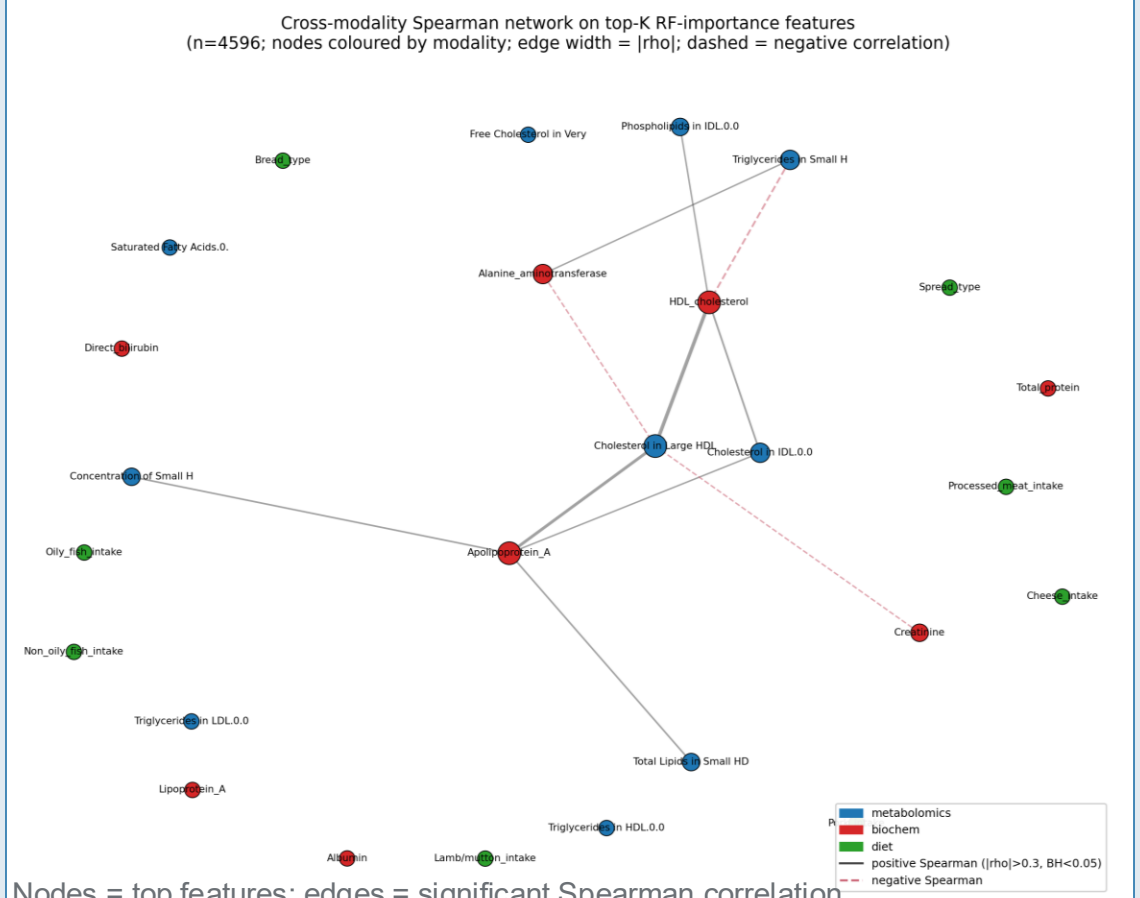




COMMON VARIABLES

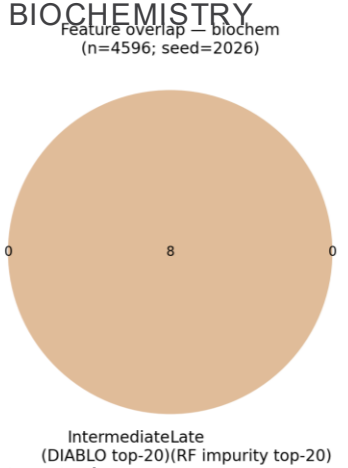
DIABLO loadings vs RF importance, and cross-modality graph.

CROSS-MODALITY NETWORK



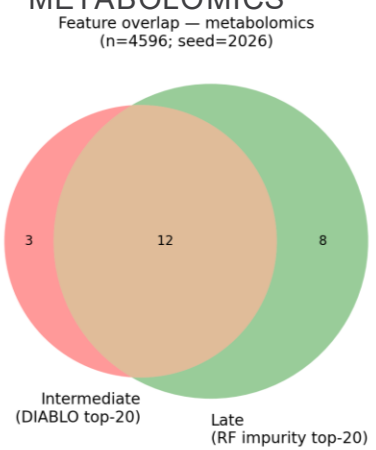
Nodes = top features; edges = significant Spearman correlation (thicker = stronger). BH-adjusted p < 0.05.

VENN · BIOCHEMISTRY



8 / 8 saturated

VENN · METABOLOMICS



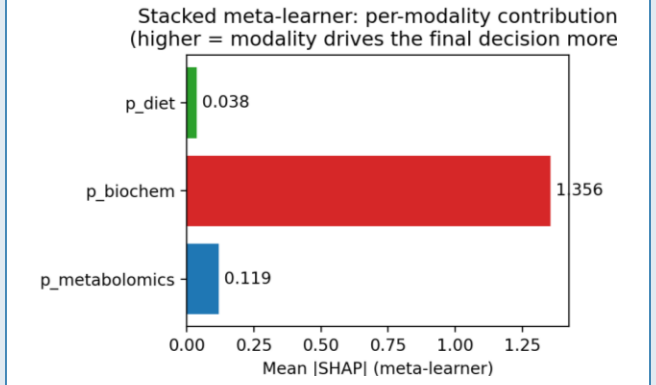
12 / 20 overlap (Jaccard 0.52)

- **Biochemistry:** 8/8 features most stable across evidence layers.
- **Metabolomics:** 12/20 features common at top 20 (Jaccard 0.52); a robust HDL / triglyceride axis.
- **Cross-modality network:** top 10 per block, Spearman $|\rho| > 0.3$, BH-adjusted $p < 0.05 \rightarrow$ 26 nodes, 11 cross-modality edges.
- **Dominant bridge:** HDL particle subfractions ↔ HDL-cholesterol & ApoA-I (ρ up to 0.83).

MODEL UNDERSTANDING

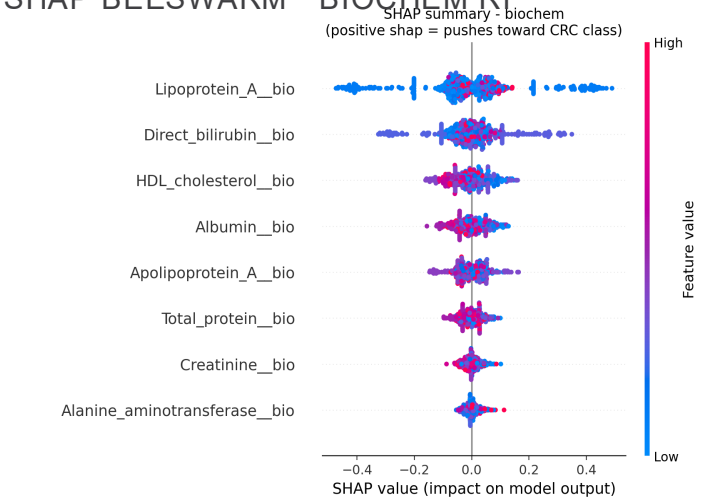
SHAP, stacked SHAP, surrogate tree, and the convergence with Discovery Track.

STACKED SHAP · PER BLOCK



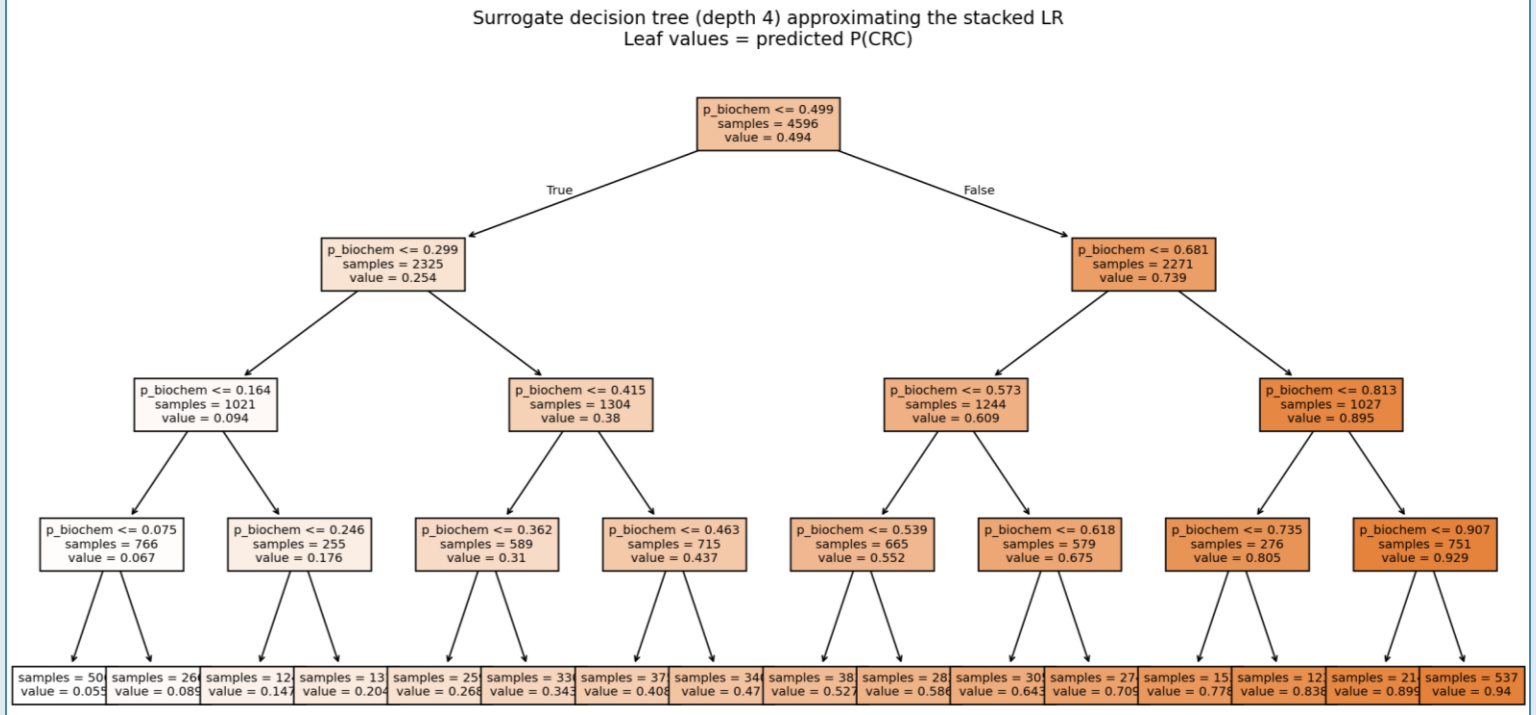
Mean |SHAP| per modality on the Stacked SHAP.

SHAP BEESWARM · BIOCHEM RF



Dot = participant; horizontal = push toward CRC; colour = feature value.

SURROGATE DECISION TREE · DEPTH 4



Approximates the stacked classifier. Every split is on p_biochem.

Stacked SHAP: the biochemistry probability dominates the fused decision.

SHAP per-RF: biochemistry's top drivers are HDL-cholesterol, Apolipoprotein-A, albumin (already routine clinical assays).

Surrogate tree (depth 4) approximating stack: every split on p_biochem; p_metabolomics and p_diet never appear. Evidence the fusion is biochem led.

Convergence: Biochemistry's SHAP drivers are the same HDL / ApoA-I axis DIABLO and rCCA recovered.

CLINICAL INTERPRETATION

Candidate markers need cautious interpretation. Every citation resolved on PubMed.

MARKER	IN CRC	PATHWAY	VERIFIED EVIDENCE
Apolipoprotein A-I	↓ decreased	HDL apoprotein, plasma proteomic biomarker	ApoA-I as prognostic biomarker in CRC[1] · ApoA-I CRC biomarker, quantitative MS + protein microarray[2]
HDL cholesterol	↓ decreased	Lipid metabolism, cholesterol transport	HDL-cholesterol inversely associated with CRC[3] meta-analysis, ~1.98M
Serum albumin	↓ decreased	Nutritional, acute-phase inflammation	Pre-operative albumin in CRC[4]
Direct bilirubin	No association	Hepatic, antioxidant (heme catabolism)	Bilirubin NULL in CRC, AMORIS cohort + meta-analysis[5] NULL in CRC

Converging evidence

DIABLO loadings + rCCA components + cross-modality network + SHAP drivers + verified literature.

Five independent strands all point at one biology: a coordinated fall in HDL-cholesterol and Apolipoprotein-A in CRC, with serum albumin adding a nutritional / inflammatory axis.

Flagged Direct Bilirubin

RF assigns it high importance; Monroy Iglesias, 2021, found *no* CRC association.

We flag the disagreement rather than silently drop it.

Feature importance is association within *this* cohort, not established biology.

ApoA-I | HDL-C | Albumin | Bilirubin | flagged

DIFFICULTIES

The items abstracted here shape interpretation

Disjoint proteomics cohort

- n = 509, **zero sample-ID** overlap, severe per-protein missingness (GLIPR1 99.8%) Excluded by data design.

Imputation uncertainty

- Lipoprotein-A (24%) and bilirubin (17%) partly imputed. Importance ranks survive, but effect magnitudes read with care.

Weak standalone modalities

- Metabolomics and diet are near-chance learners (AUC 0.57 / 0.51); Boruta confirmed 0/50 metabolite and 0/8 diet features at full strictness.

DIABLO under-predicts

- BER 0.43 ≈ AUC 0.59, a sparse linear projection cannot match RF's non-linear interactions. Its value is discovery, not prediction.

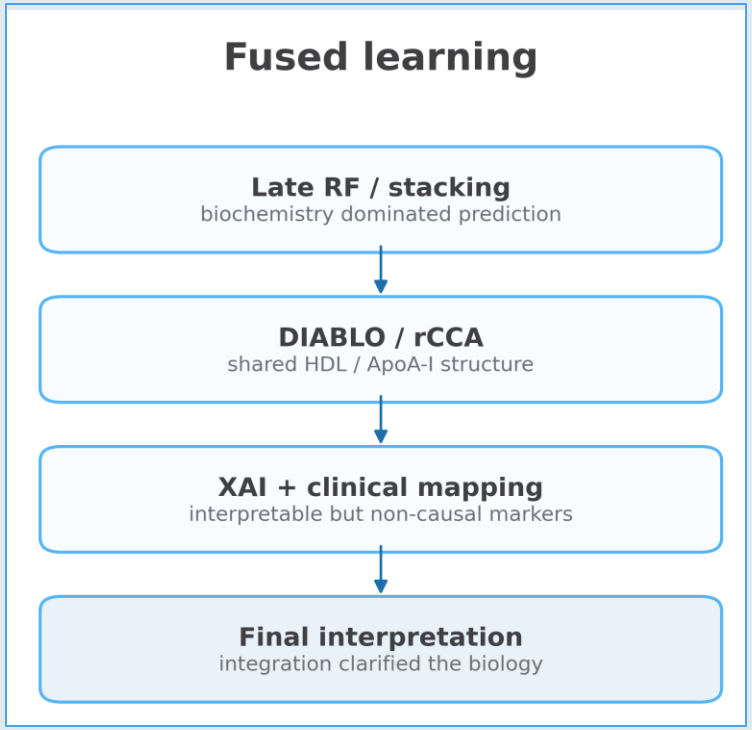
Cross-sectional, binary label

- No stage, MSI status or tumor location; associations only. No causal or subtype-identification claims.

Fusion gain is small

- Late fusion buys only +0.0034 AUC. Integration deliverable is reported.

CONCLUSION



Summary

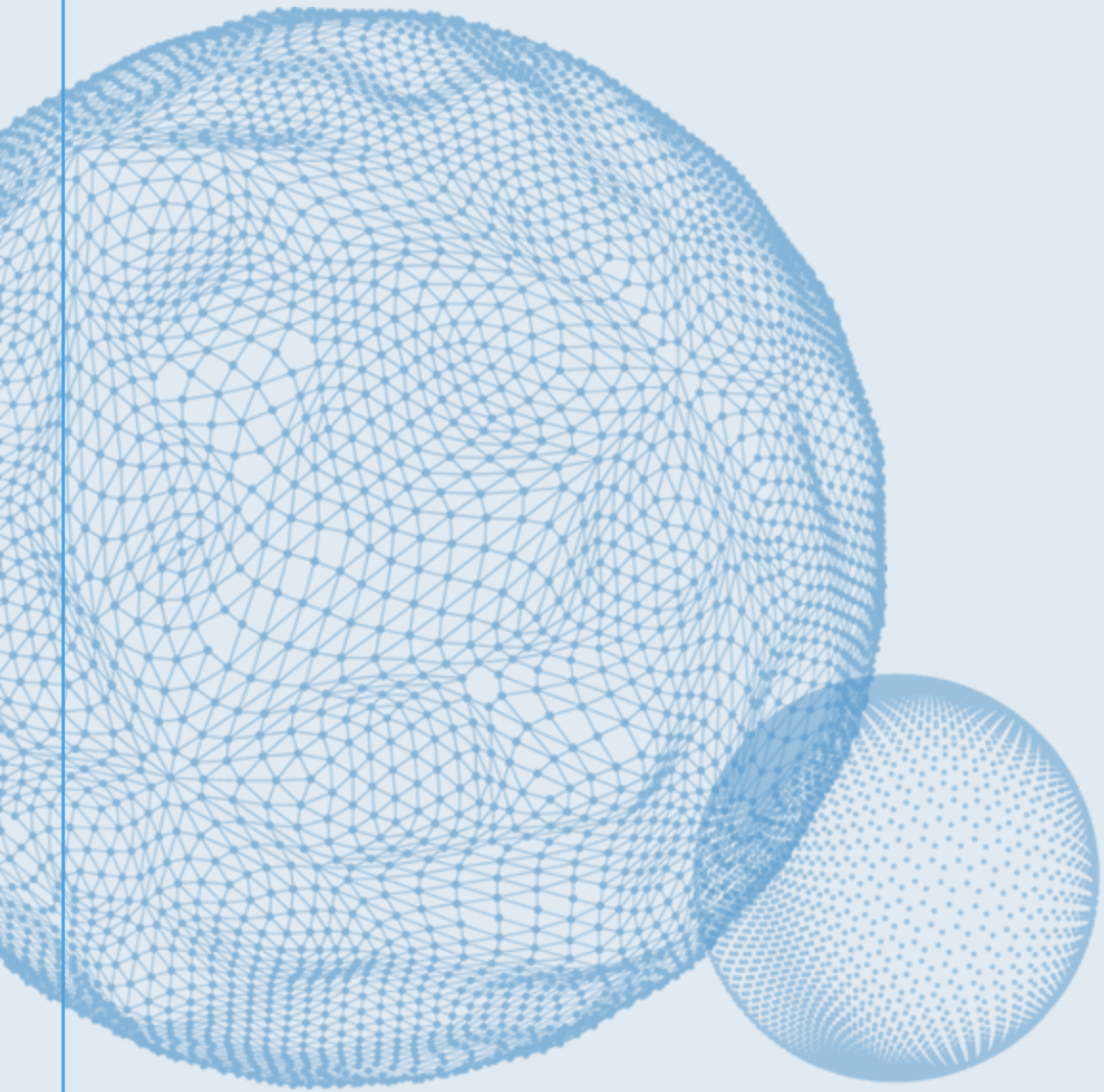
- Biochemistry was the dominant CRC predictive modality (AUC 0.824).
- Stacked fusion gave only a small AUC gain (+0.0034).
- Intermediate integration revealed coherent HDL / ApoA-I and hepatic axes (ρ = 0.884).
- XAI confirmed that the model was biochemistry-led.
- Candidate biomarkers require external validation.

Verdict

- The main value of integration was not dramatic prediction improvement, but a clearer biological explanation of a biochemistry-led CRC signal.

Future Direction

Patient-matched proteomics on the same cohort → a true four-block integration.
Longitudinal sampling for temporal/causal direction.
Staged outcomes → move from biomarker discovery toward subtype identification.



THANK YOU

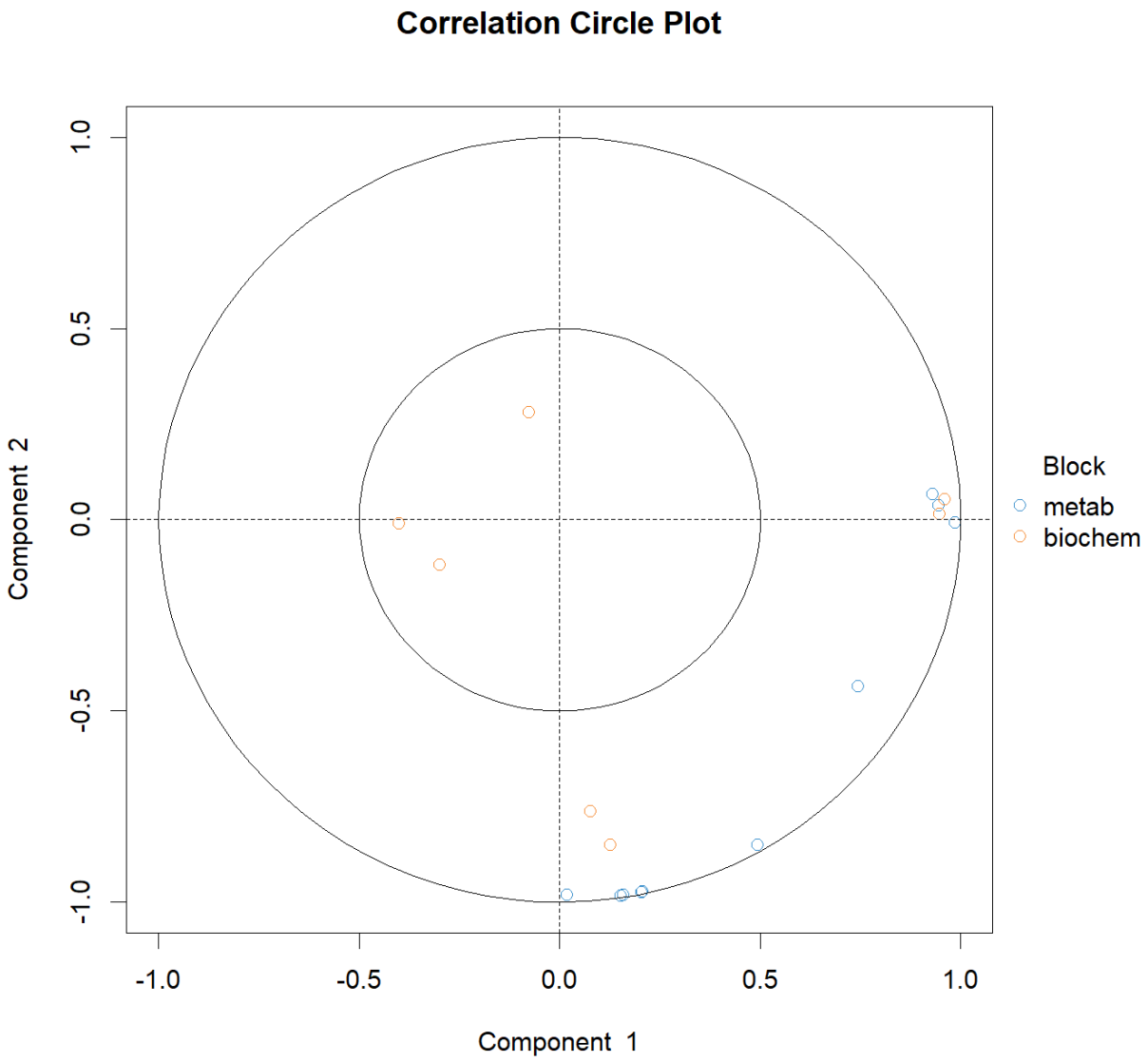
RIYA DEEPAK SHET

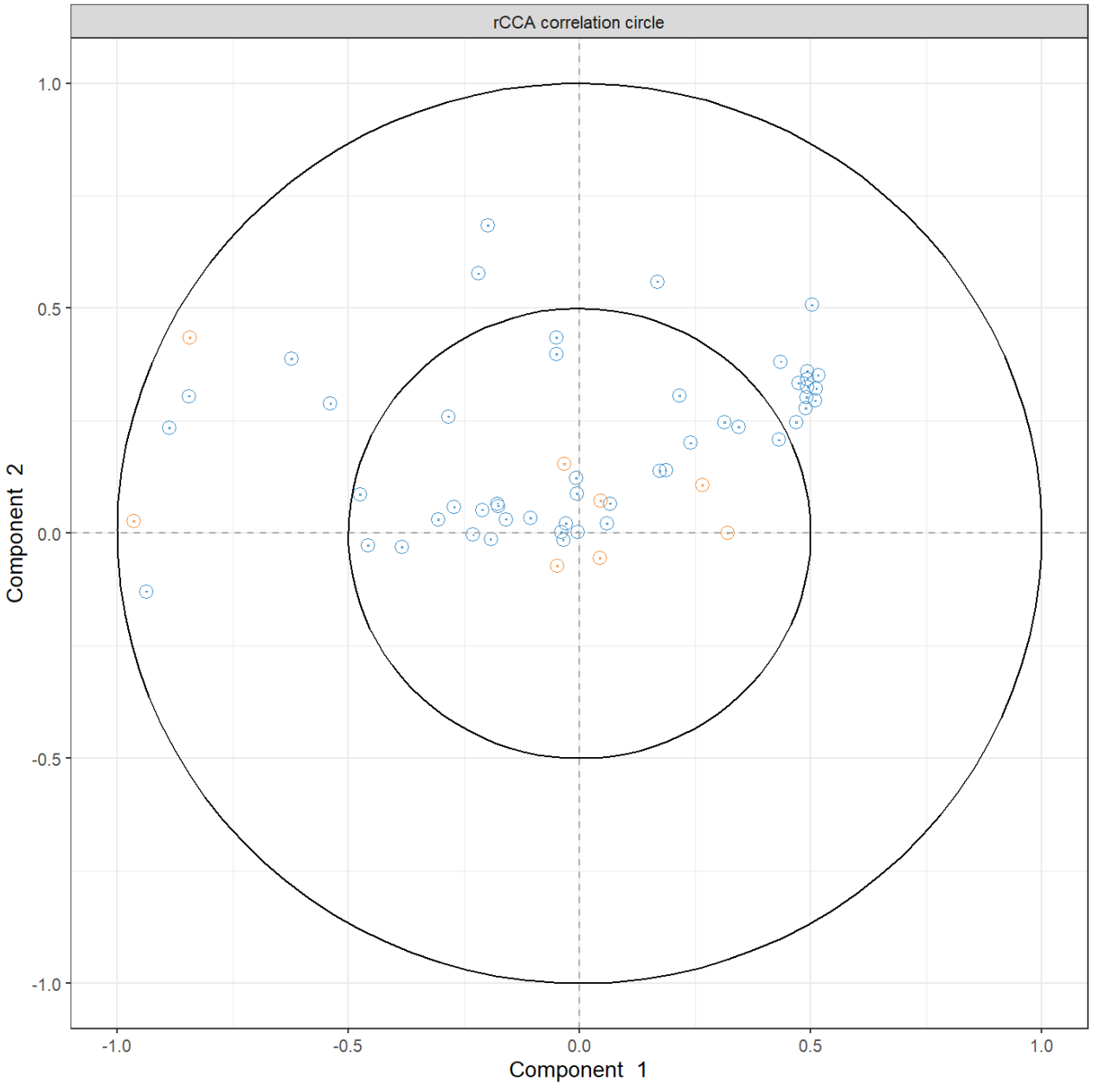
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10. Filzmoser, P., Maronna, R. and Werner, M. (2007) Outlier identification in high dimensions. *Computational Statistics & Data Analysis*, 52 (3): 1694–1711. doi:10.1016/j.csda.2007.05.018.

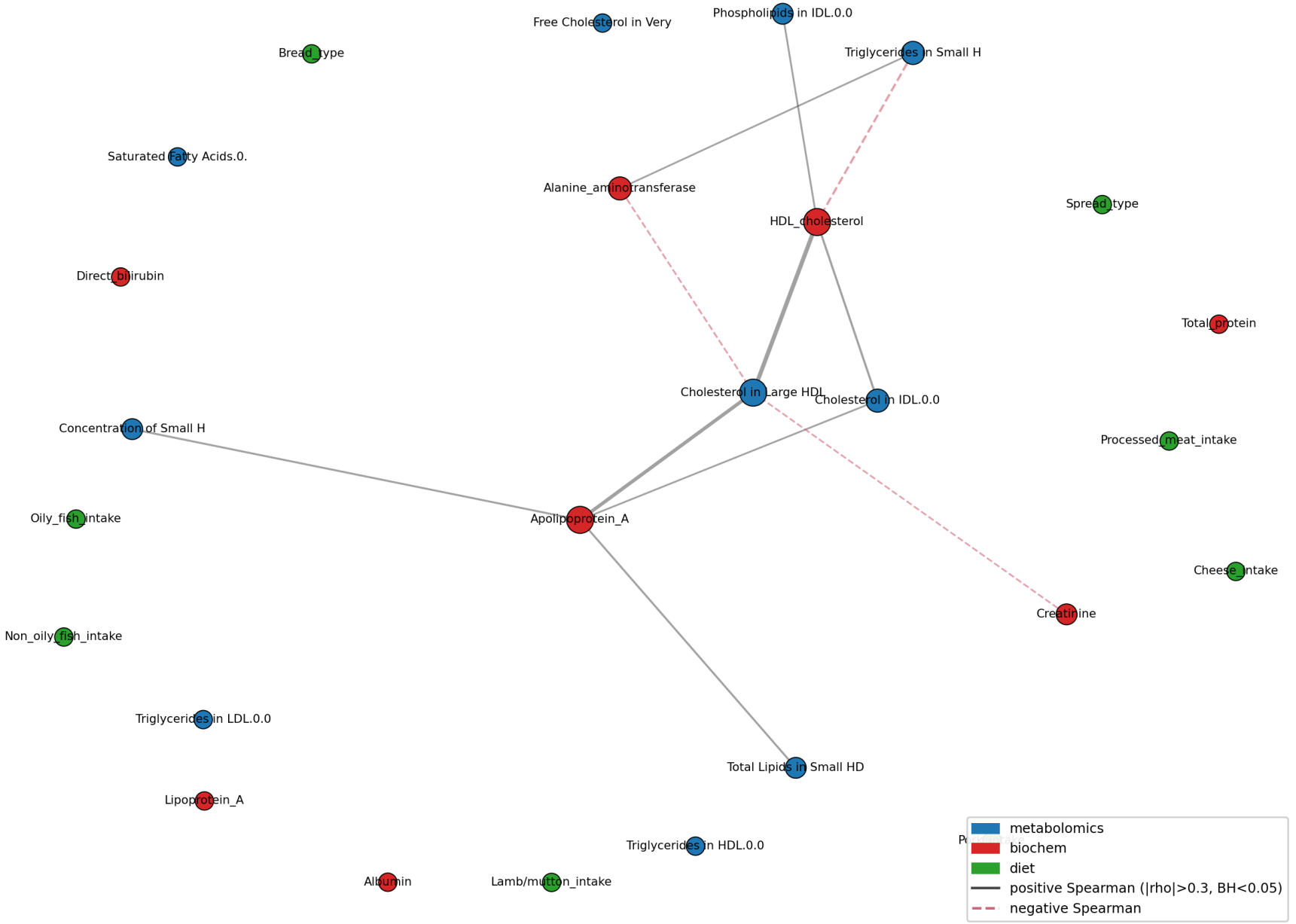
BACKUP SLIDES

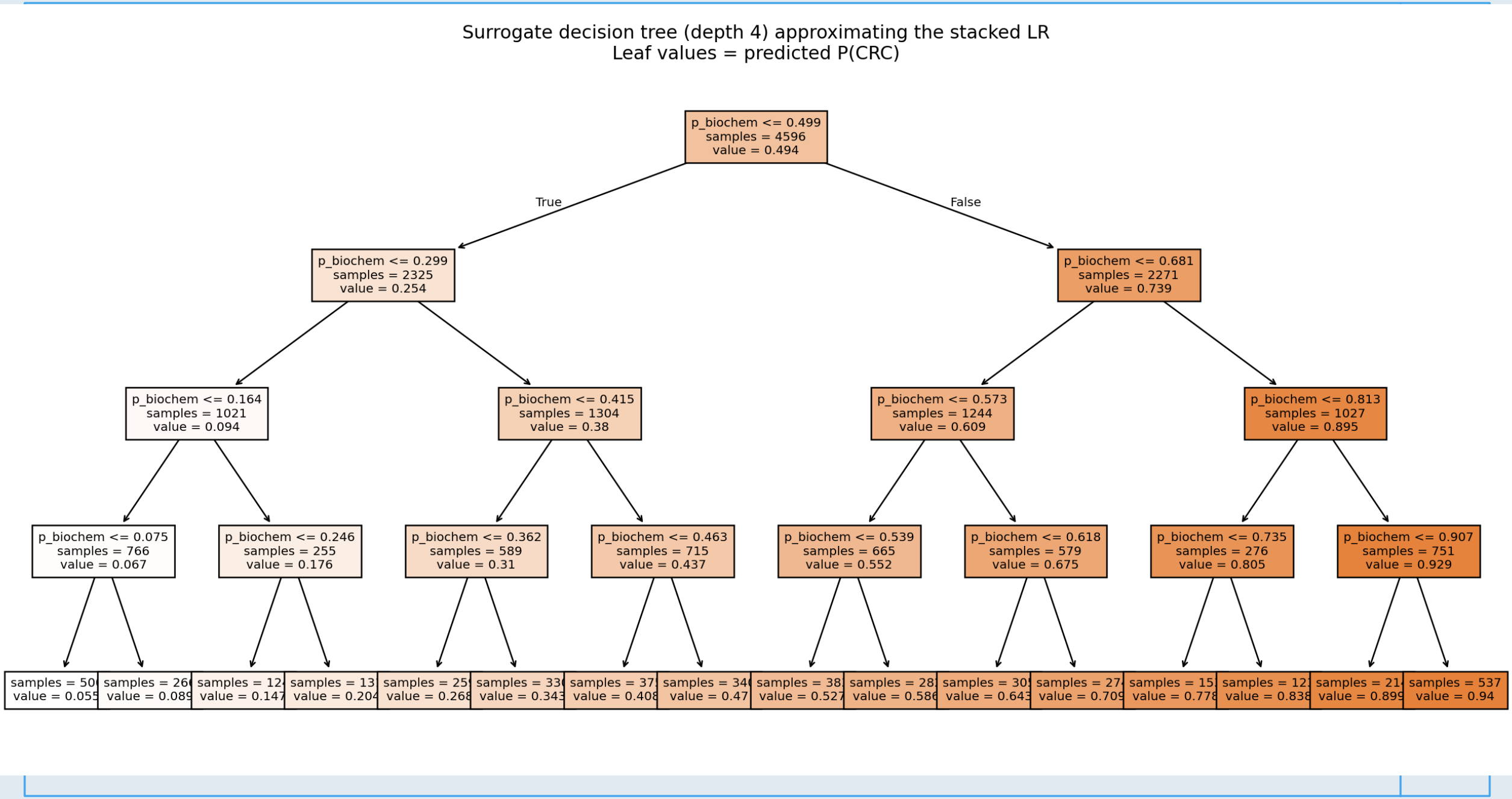
ENLARGED IMAGES OF THE FIGURES, FOR CLARITY AND TRANSPARENCY


Correlation Circle Plot
Component 2
Component 1
1.0
0.5
0.0
-0.5
-1.0
Block
metab
biochem

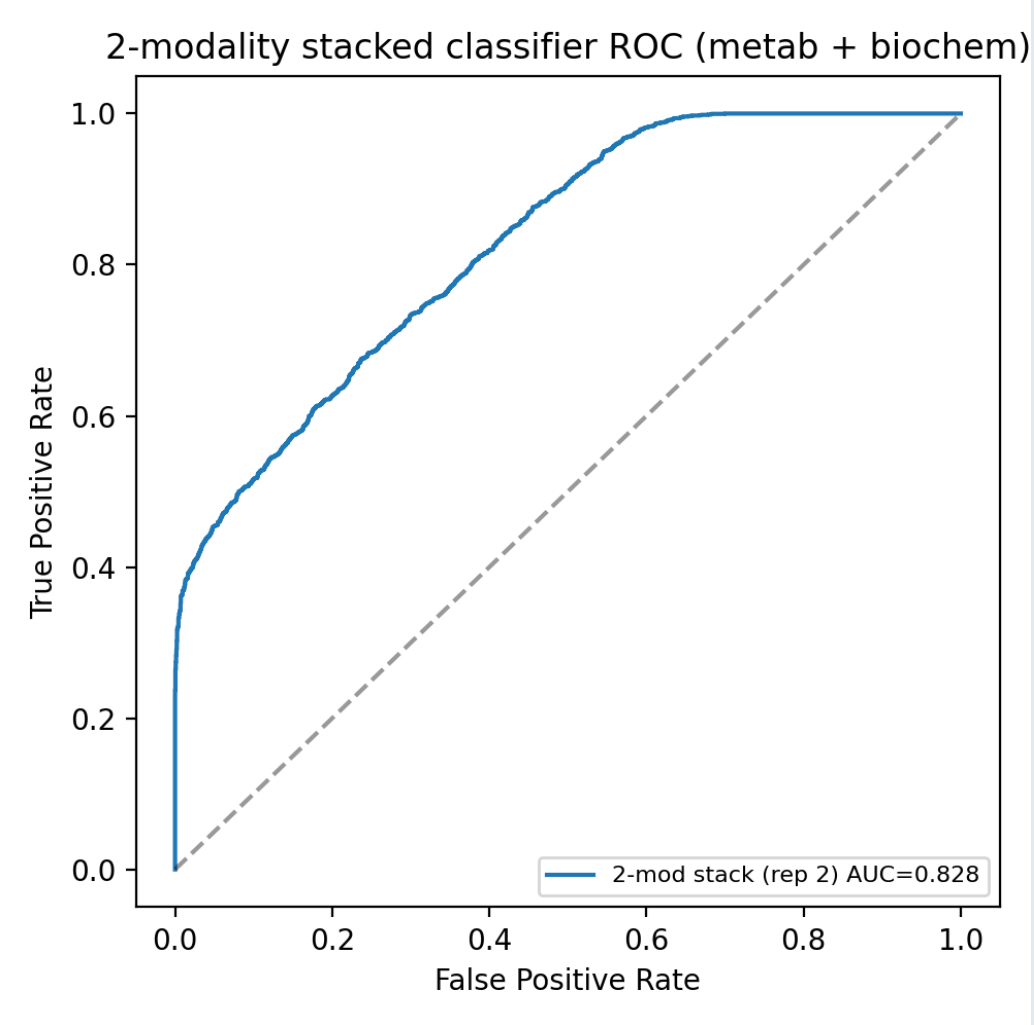

rCCA correlation circle
Component 2
Component 1
1.0
0.5
0.0
-0.5
-1.0

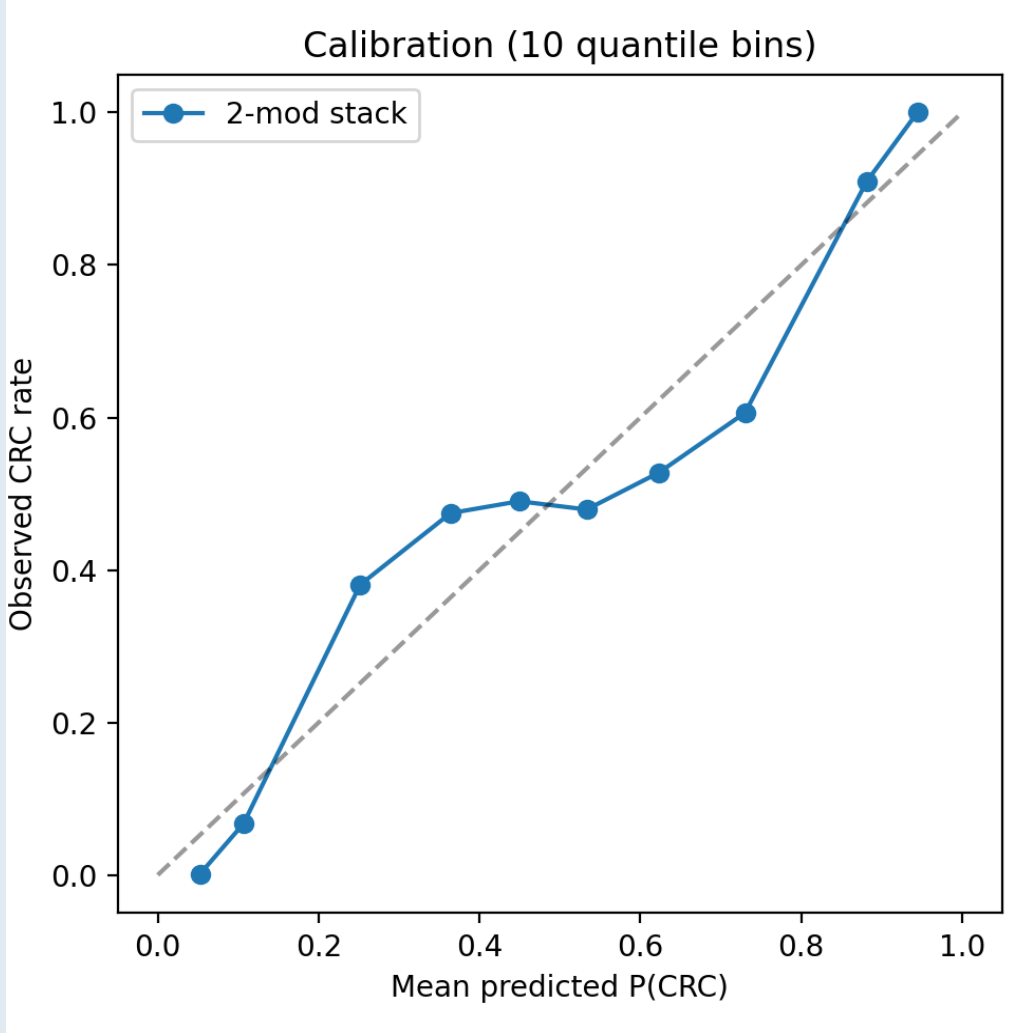
Cross-modality Spearman network on top-K RF-importance features
(n=4596; nodes coloured by modality; edge width = |rho|; dashed = negative correlation)



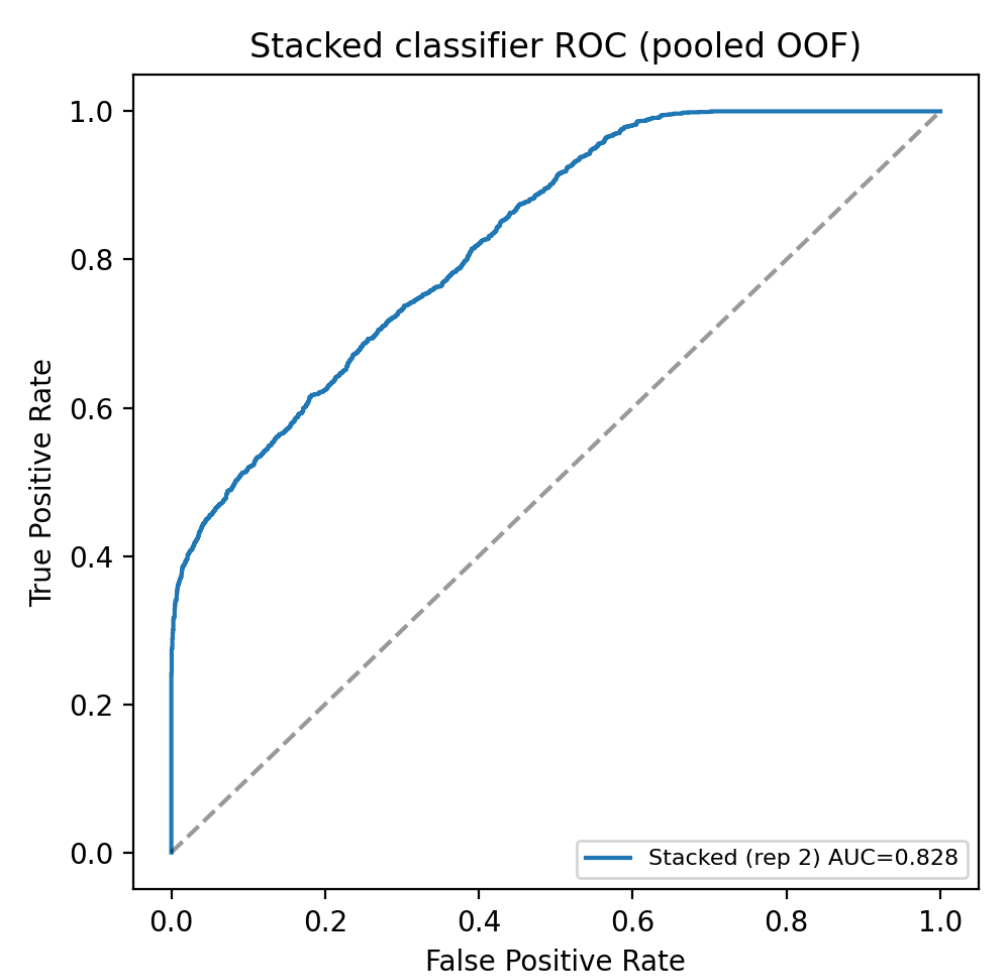

Surrogate decision tree (depth 4) approximating the stacked LR
Leaf values = predicted P(CRC)
p_biochem <= 0.499
samples = 4596
value = 0.494
True
False
p_biochem <= 0.299
samples = 2325
value = 0.254
p_biochem <= 0.681
samples = 2271
value = 0.739
p_biochem <= 0.164
samples = 1021
value = 0.094
p_biochem <= 0.415
samples = 1304
value = 0.38
p_biochem <= 0.573
samples = 1244
value = 0.609
p_biochem <= 0.813
samples = 1027
value = 0.895
p_biochem <= 0.075
samples = 766
value = 0.067
p_biochem <= 0.246
samples = 255
value = 0.176
p_biochem <= 0.362
samples = 589
value = 0.31
p_biochem <= 0.463
samples = 715
value = 0.437
p_biochem <= 0.539
samples = 665
value = 0.552
p_biochem <= 0.618
samples = 579
value = 0.675
p_biochem <= 0.735
samples = 276
value = 0.805
p_biochem <= 0.907
samples = 751
value = 0.929
samples = 50
value = 0.055
samples = 26
value = 0.089
samples = 12
value = 0.147
samples = 13
value = 0.204
samples = 25
value = 0.268
samples = 33
value = 0.343
samples = 37
value = 0.408
samples = 34
value = 0.47
samples = 38
value = 0.527
samples = 28
value = 0.586
samples = 30
value = 0.643
samples = 27
value = 0.709
samples = 15
value = 0.778
samples = 12
value = 0.838
samples = 21
value = 0.899
samples = 537
value = 0.94

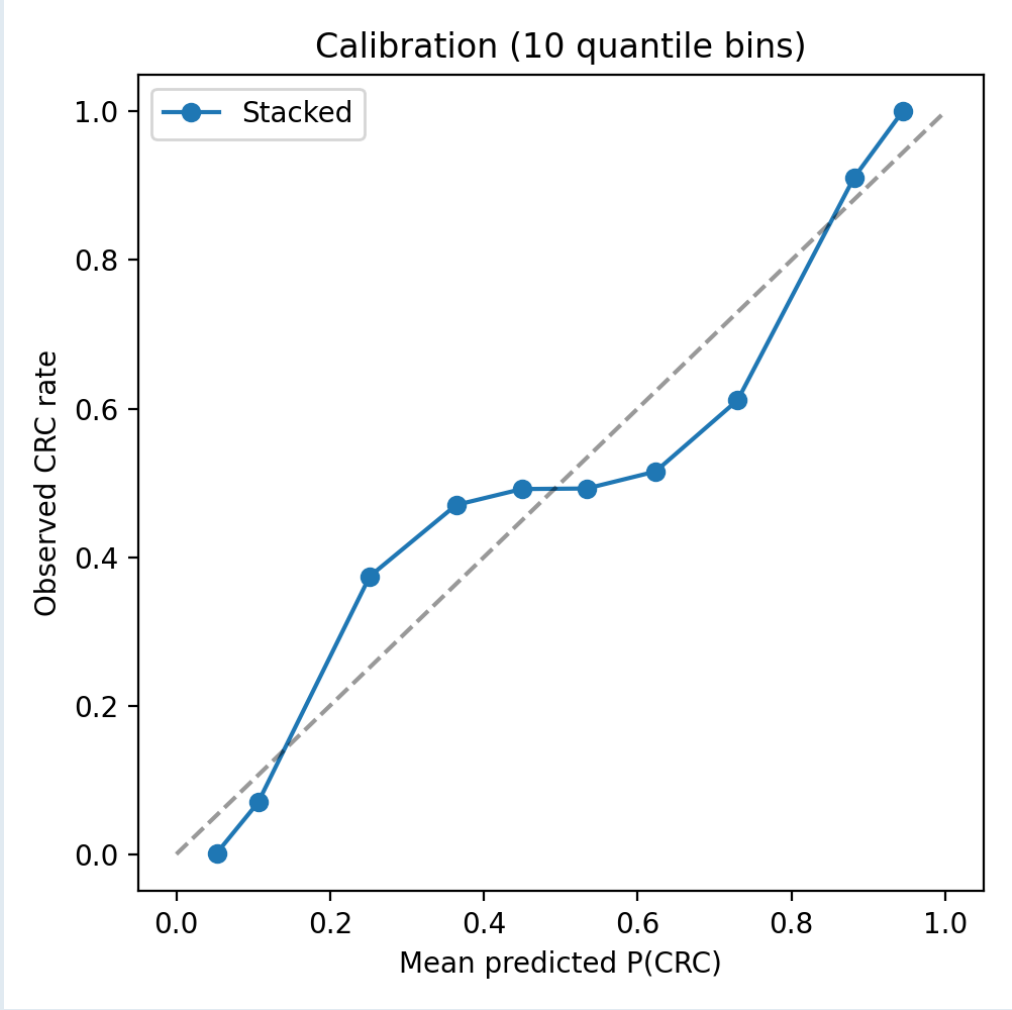
ROC (2-MODALITY STACK) AND CALIBRATION OF THE FUSED CLASSIFIER.



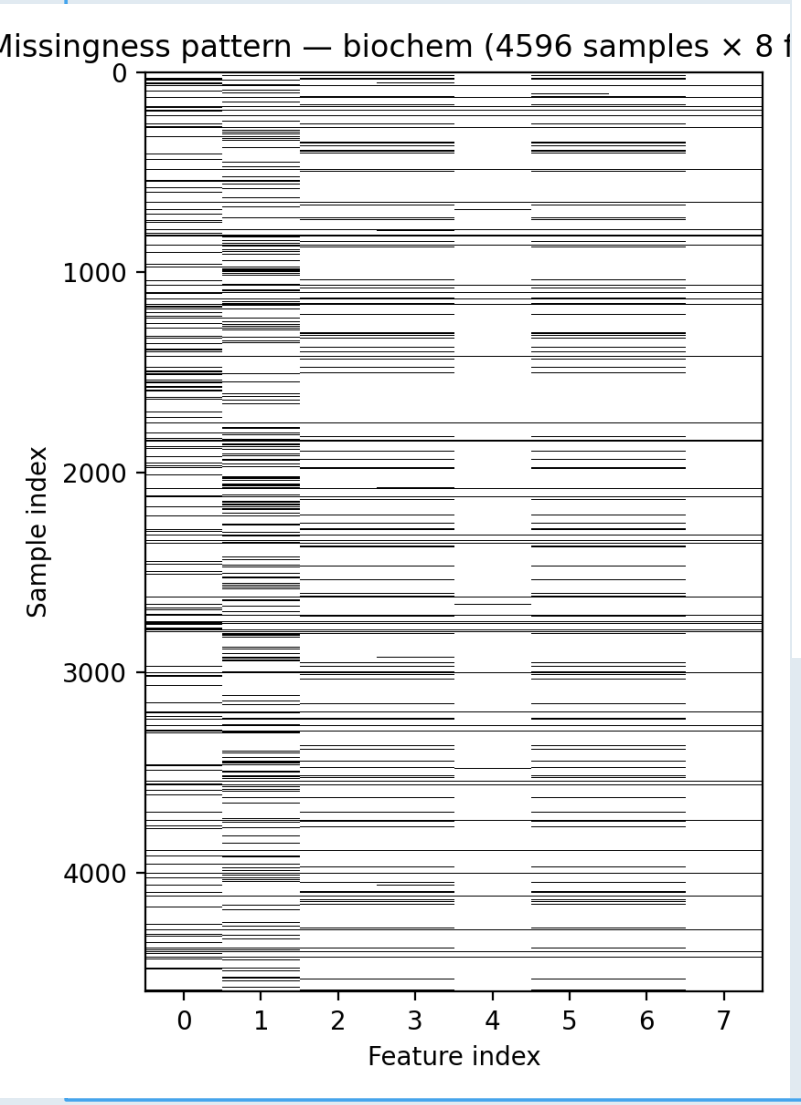


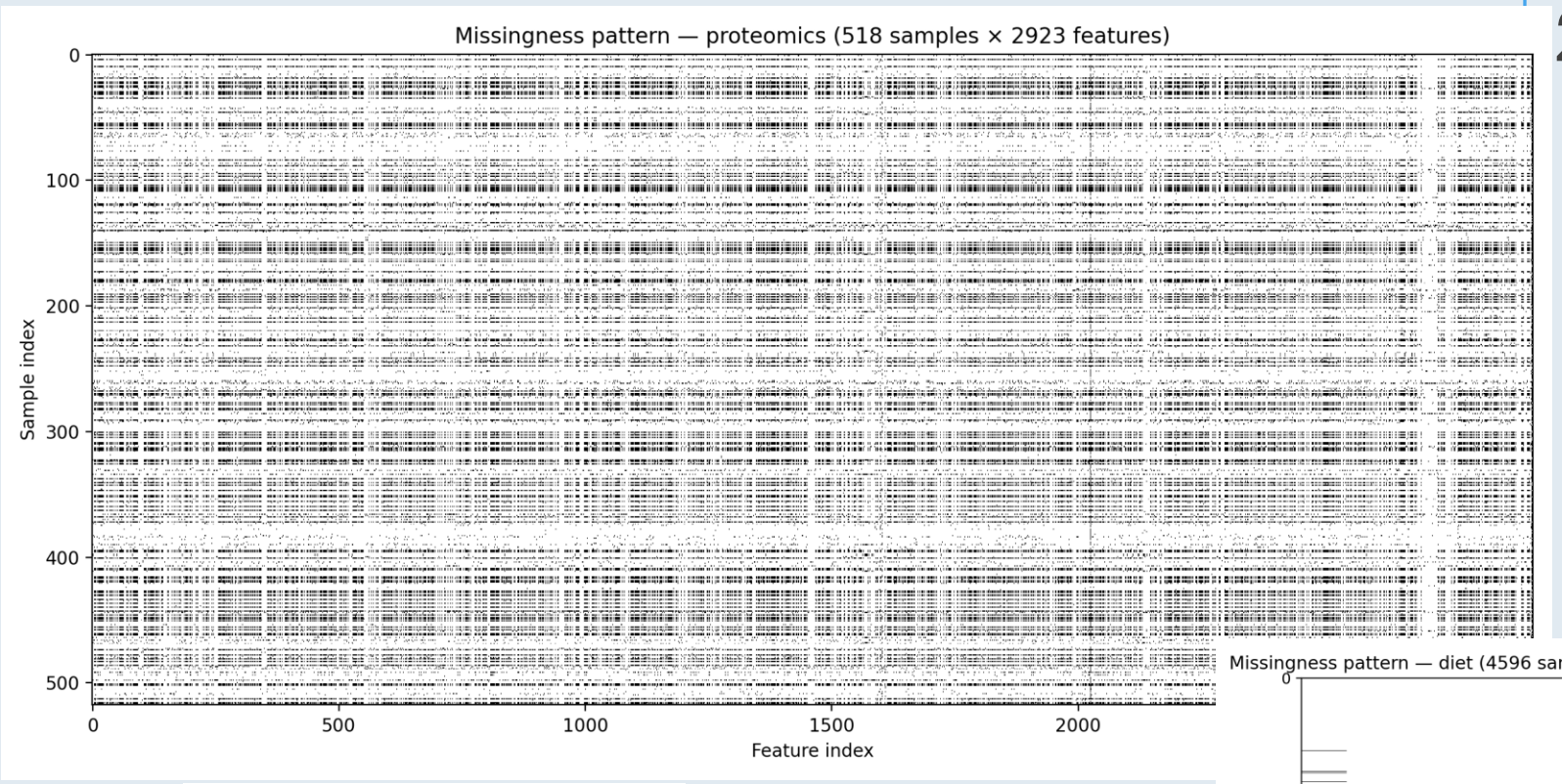
ROC AND CALIBRATION (INCL. DIET RF)

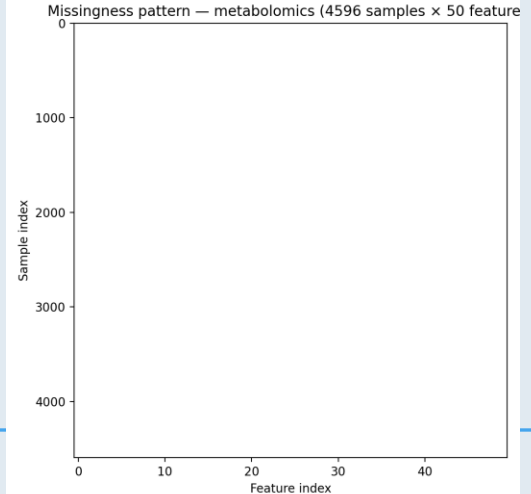


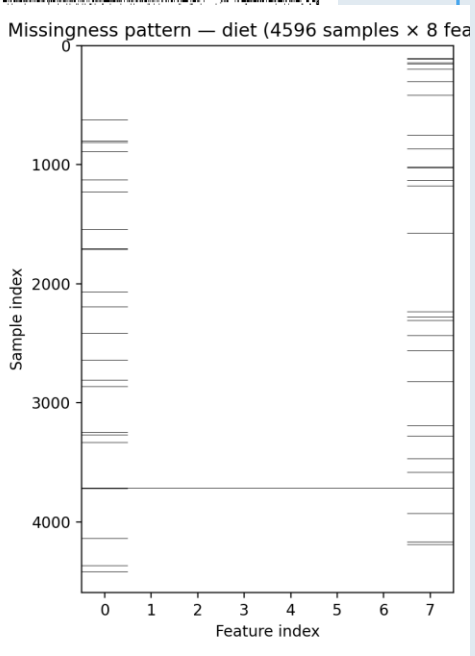


MISSINGNESS PATTERNS- ALL MODALITIES

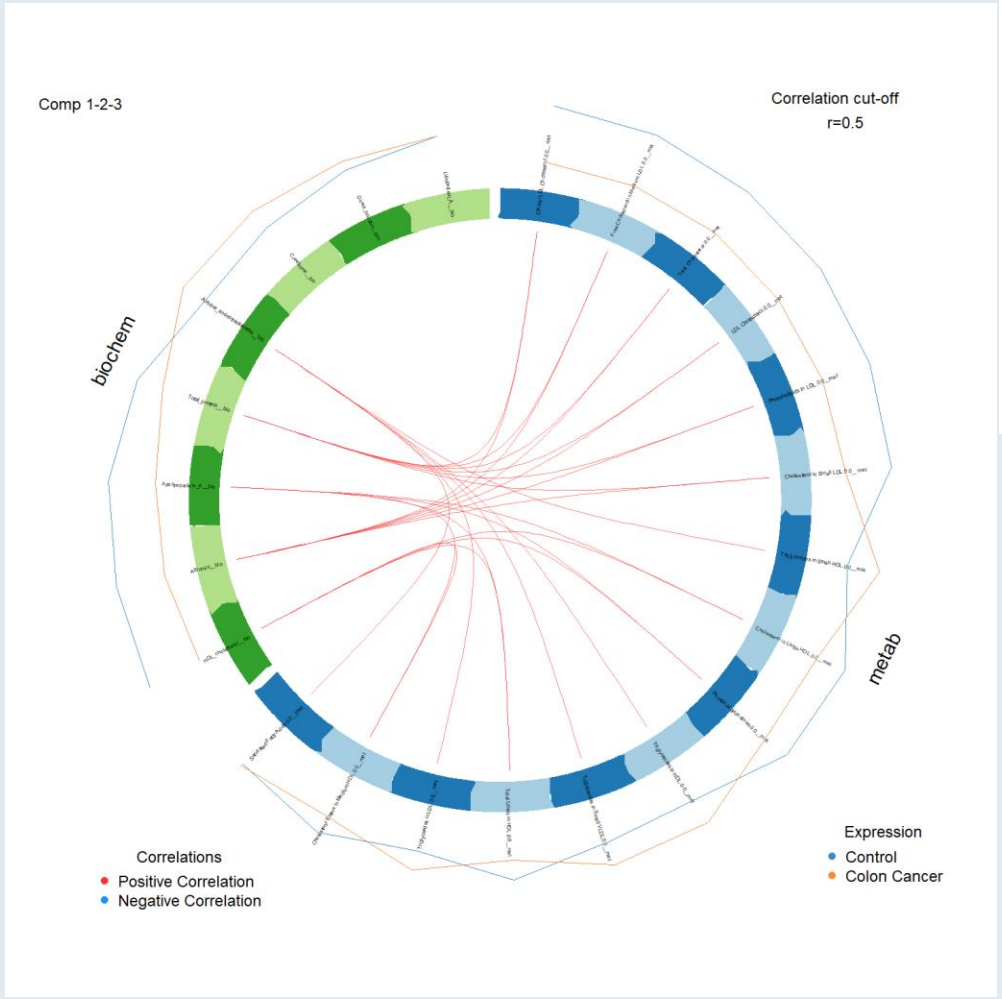


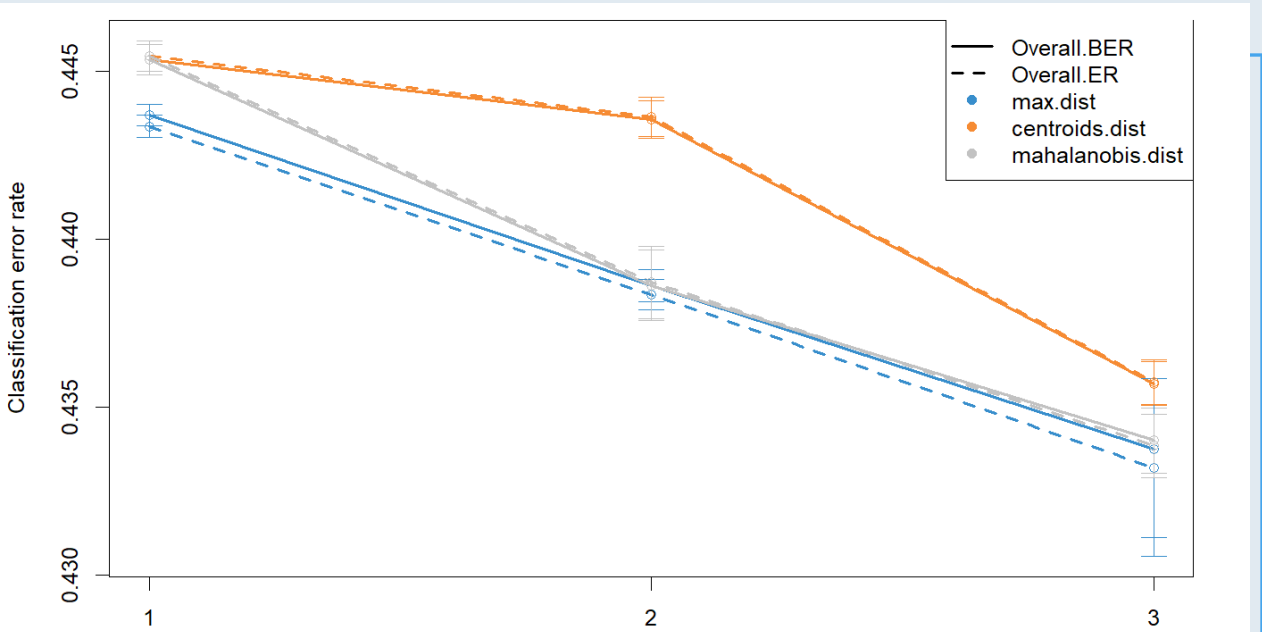


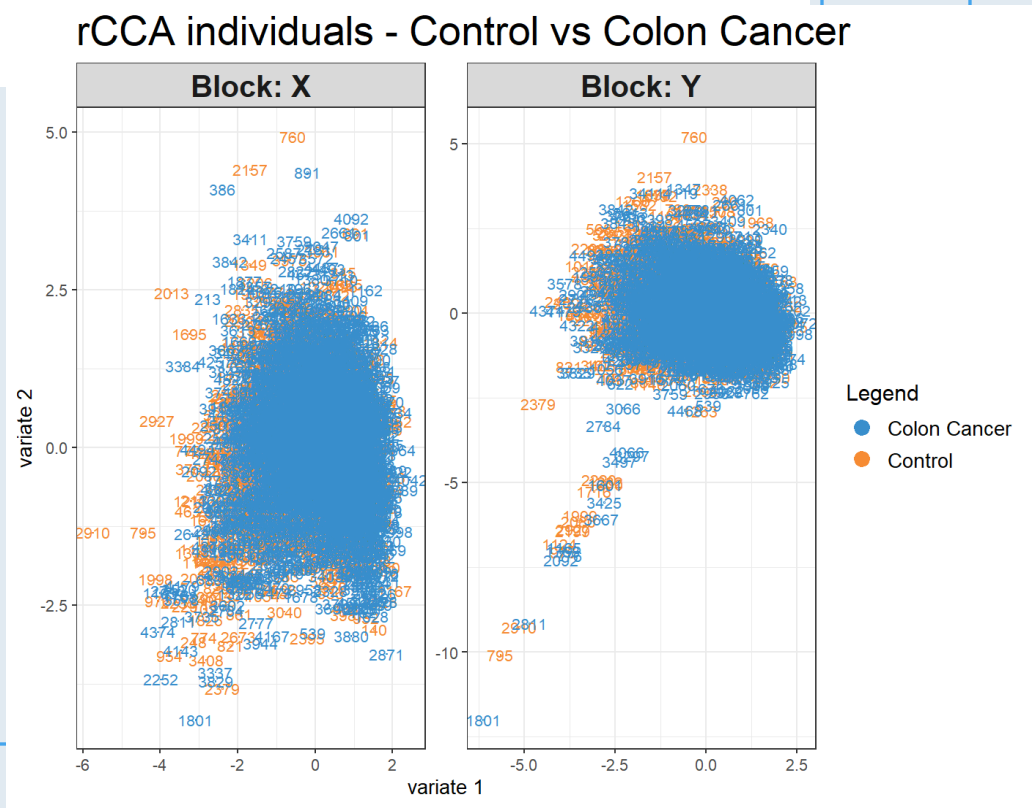




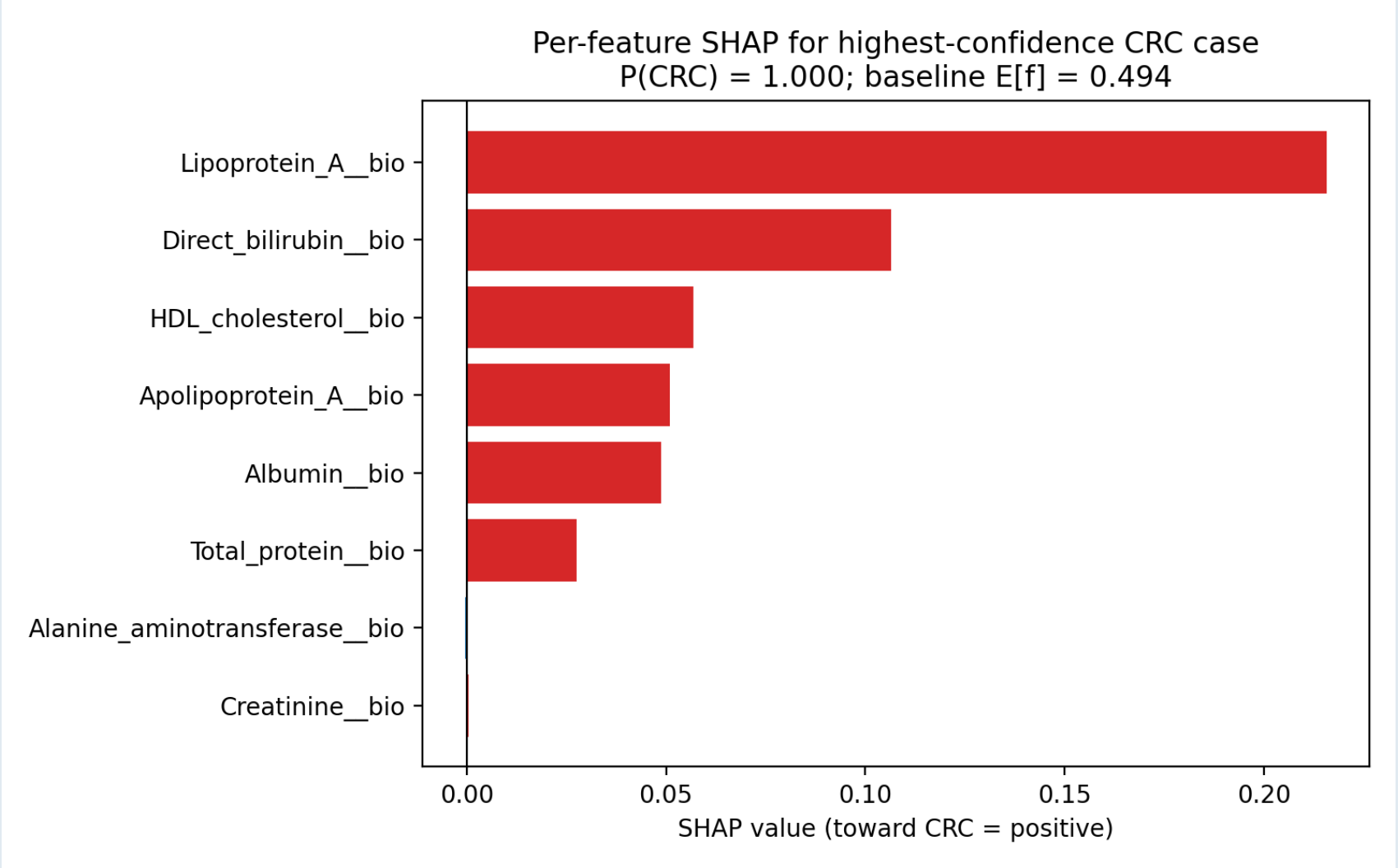
DIABLO AND RCCA







SHAP PER-FEATURE CONFIDENCE



SHAP BEESWARM PER MODALITY

